Customer Analytics Dashboard using Machine Learning

Project Report

Abstract

This project focuses on developing an interactive Customer Analytics Dashboard using Python, Machine Learning, and Streamlit to analyze customer behavior and generate business insights. The system applies RFM analysis, K-Means clustering, Customer Lifetime Value analysis, churn risk identification, and AI-powered recommendations to help businesses improve customer engagement and revenue optimization.

Introduction

Customer analytics helps organizations understand customer purchasing behavior and improve marketing strategies. This project was developed to provide an intelligent analytics platform capable of customer segmentation and business insight generation through interactive visualizations and machine learning techniques.

Objectives

- Perform customer segmentation using RFM analysis and K-Means clustering.
- Analyze customer purchasing behavior and spending patterns.
- Build an interactive dashboard using Streamlit.
- Generate AI-powered business insights using Groq API.
- Deploy the project online using Streamlit Cloud.

Technologies Used

- Python
- Pandas and NumPy
- Scikit-learn
- Streamlit
- Plotly
- Matplotlib and Seaborn
- Groq API
- GitHub and Streamlit Cloud

Dataset Description

The project uses the Online Retail dataset containing customer transaction records such as invoice details, quantities, prices, customer IDs, and countries. Additional features including Recency, Frequency, Monetary value, Customer Lifetime Value, and customer personas were generated during preprocessing.

Methodology

- Data cleaning and preprocessing
- RFM feature engineering
- Data normalization using StandardScaler
- Customer segmentation using K-Means clustering
- Dashboard development using Streamlit
- AI-powered recommendation generation

RFM Analysis

RFM analysis evaluates customer behavior based on Recency, Frequency, and Monetary value. Customers with recent purchases, higher purchase frequency, and greater spending are considered more valuable for businesses.

Customer Segmentation

K-Means clustering was used to divide customers into meaningful groups such as VIP Customers, Loyal Customers, New Customers, and At-Risk Customers. These segments help businesses personalize marketing strategies and improve customer retention.

Dashboard Features

- Dark-themed interactive dashboard
- KPI cards for customer analytics
- Interactive filters for persona, country, and revenue
- 3D customer segmentation visualization
- Correlation heatmap
- Top customer analysis
- Downloadable reports
- AI-generated business insights

AI-Powered Insights

The project integrates Groq API to generate automated business insights and recommendations. The AI module analyzes customer metrics and provides suggestions related to customer retention, marketing strategies, churn reduction, and revenue optimization.

Results and Observations

- VIP customers contribute the highest revenue.
- Loyal customers show high purchase frequency.
- At-risk customers require re-engagement strategies.
- Interactive filtering improves analytical flexibility.
- AI insights enhance business decision-making.

Advantages of the System

- Interactive and user-friendly interface
- Real-time customer analysis
- Improved customer segmentation
- Business intelligence support
- Cloud-based deployment accessibility

Future Enhancements

- Predictive churn modeling
- Recommendation systems
- Database integration
- Natural language query support
- Mobile-responsive interface
- Automated PDF report generation

Conclusion

The Customer Analytics Dashboard successfully demonstrates the integration of Machine Learning, Data Visualization, and AI-powered analytics for customer behavior analysis. The system provides valuable business insights and supports data-driven decision-making through an interactive and scalable analytics platform.